



PicoMagnetometer RPi software User manual

Pico^{nT}

<https://ukraa.com/>

The UK Radio Astronomy Association

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Python code for the UKRAA PicoMagnetometer

Set of Python, gnuplot and shell scripts to run on an RPi4/5 to record, process and present data from the UKRAA PicoMagnetometer.

This software was written to suit a specific set-up; feel free to use as you see fit.

Instructions for initial setting up of a Raspberry Pi4/5 are included in the **docs** folder

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Using the code

The code assumes that your UKRAA PicoMagnetometer is the only device using USB that is connected to the RPi4/5 via supplied USB cable and that it is **/dev/ttyACM0** - you can check this by using **ls /dev/tty*** in a terminal window on the RPi4/5 and reviewing the response.

The code assumes username is **pi**.

If **pi** is not the username, then you will need to change all occurrences of **'/home/pi'** to **'/home/username'** in the python, gnuplot and shell scripts, where **username** is the username you have selected for your RPi4/5.

The code assumes one magnetometer connected to the RPi4/5 USB and that it will be connected via **/dev/ttyACM0**.

If there are other devices connected to the RPi and your magnetometer is not **/dev/ttyACM0**, then you will need to change **/dev/ttyACM0** to **/dev/ttyACMx** in the **GetDataRawACM0.py** python script, where **ttyACMx** is the tty address of you connected magnetometer.

GetDataRawACM0.py is run as a service.

Other scripts (Python and gnuplot) are run from **cron** using shell scripts.

Where is my PicoMagnetometer?

Plug your PicoMagnetometer into any of the RPi USB ports - I normally use one of the blue ports (USB3).

1. Open a terminal window and type the following command and press enter

ls /dev/tty*



```
pi@TEST-MAG: ~  
File Edit Tabs Help  
pi@TEST-MAG:~$ ls /dev/tty*  
/dev/tty /dev/tty19 /dev/tty3 /dev/tty40 /dev/tty51 /dev/tty62  
/dev/tty0 /dev/tty2 /dev/tty30 /dev/tty41 /dev/tty52 /dev/tty63  
/dev/tty1 /dev/tty20 /dev/tty31 /dev/tty42 /dev/tty53 /dev/tty7  
/dev/tty10 /dev/tty21 /dev/tty32 /dev/tty43 /dev/tty54 /dev/tty8  
/dev/tty11 /dev/tty22 /dev/tty33 /dev/tty44 /dev/tty55 /dev/tty9  
/dev/tty12 /dev/tty23 /dev/tty34 /dev/tty45 /dev/tty56 /dev/ttyACM0  
/dev/tty13 /dev/tty24 /dev/tty35 /dev/tty46 /dev/tty57 /dev/ttyprintk  
/dev/tty14 /dev/tty25 /dev/tty36 /dev/tty47 /dev/tty58  
/dev/tty15 /dev/tty26 /dev/tty37 /dev/tty48 /dev/tty59  
/dev/tty16 /dev/tty27 /dev/tty38 /dev/tty49 /dev/tty6  
/dev/tty17 /dev/tty28 /dev/tty39 /dev/tty5 /dev/tty60  
/dev/tty18 /dev/tty29 /dev/tty4 /dev/tty50 /dev/tty61  
pi@TEST-MAG:~$
```

2. You are looking for **/dev/ttyACM0** - this is on the right hand side of the screen shot above.
3. This is the USB address for your attached PicoMagnetometer - if you have more than one magnetometer attached you may see **/dev/ttyACM1** etc.
4. If you do not see **/dev/ttyACM0**, then unplug and plug the PicoMagnetometer back in and try again.
5. As long as you see **/dev/ttyACM0** then you do not have to make any changes to the python scripts, because they are looking for **ACM0**.

Getting the software onto your RPi

1. Log into your Raspberry Pi4/5 using VNC.
2. Open a terminal window, type the following command and press enter

git clone https://github.com/UKradioastro/UKRAA_Magnetometer.git



```
pi@TEST-MAG: ~
File Edit Tabs Help
pi@TEST-MAG:~ $ git clone https://github.com/UKradioastro/UKRAA_Magnetometer.git
```

This will download all of the code to the directory **UKRAA_Magnetometer** inside **/home/pi**

Installing the software onto your RPi

1. Open a terminal window and type the following command and press **enter**

cd ~/UKRAA_Magnetometer/install

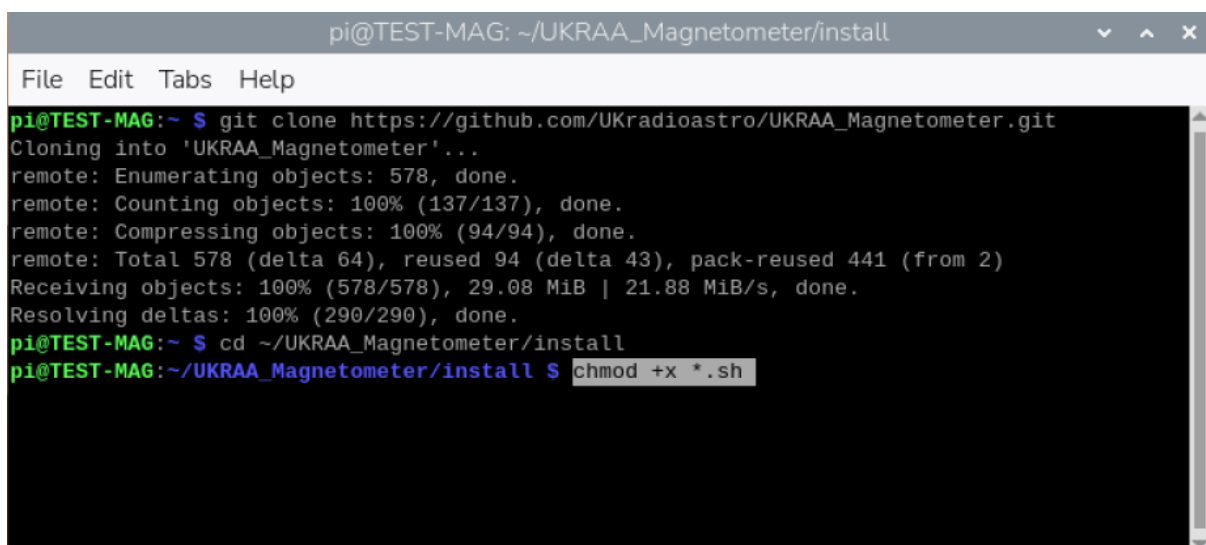
A terminal window titled 'pi@TEST-MAG: ~' with a menu bar (File, Edit, Tabs, Help). The terminal shows the output of a 'git clone' command, including progress bars for enumerating, counting, and compressing objects, and a final 'cd' command to navigate to the 'install' directory.

```
pi@TEST-MAG: ~ $ git clone https://github.com/UKradioastro/UKRAA_Magnetometer.git
Cloning into 'UKRAA_Magnetometer'...
remote: Enumerating objects: 578, done.
remote: Counting objects: 100% (137/137), done.
remote: Compressing objects: 100% (94/94), done.
remote: Total 578 (delta 64), reused 94 (delta 43), pack-reused 441 (from 2)
Receiving objects: 100% (578/578), 29.08 MiB | 21.88 MiB/s, done.
Resolving deltas: 100% (290/290), done.
pi@TEST-MAG: ~ $ cd ~/UKRAA_Magnetometer/install
```

This will take you to the **install** directory inside **/home/pi/UKRAA_Magnetometer**

2. Type the following command and press **enter**

chmod +x *.sh

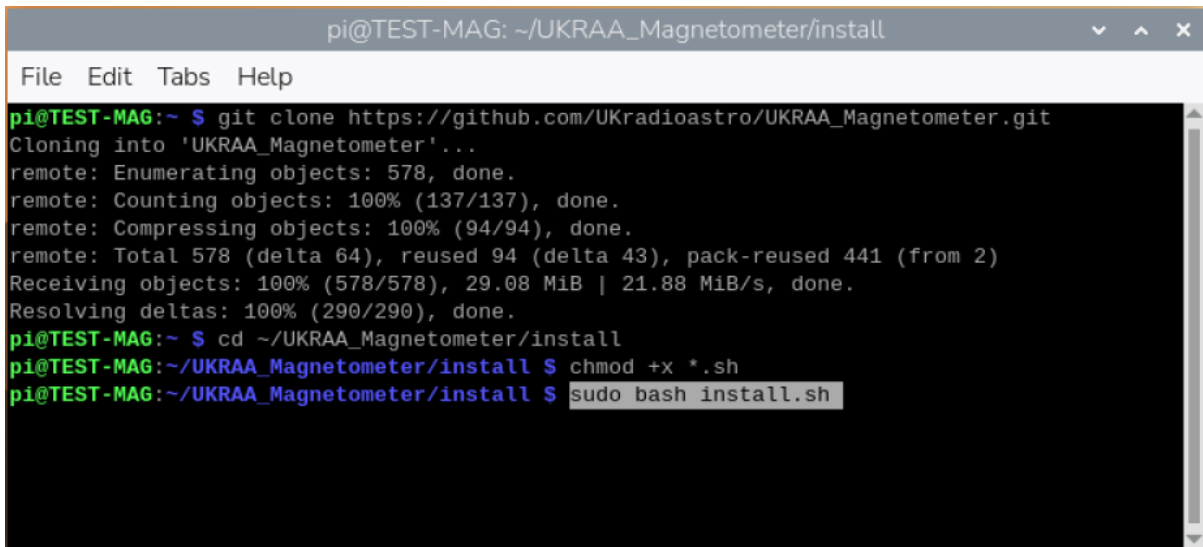
A terminal window titled 'pi@TEST-MAG: ~/UKRAA_Magnetometer/install' with a menu bar (File, Edit, Tabs, Help). The terminal shows the same 'git clone' output as the previous screenshot, followed by the 'cd' command and then the 'chmod +x *.sh' command being entered.

```
pi@TEST-MAG: ~ $ git clone https://github.com/UKradioastro/UKRAA_Magnetometer.git
Cloning into 'UKRAA_Magnetometer'...
remote: Enumerating objects: 578, done.
remote: Counting objects: 100% (137/137), done.
remote: Compressing objects: 100% (94/94), done.
remote: Total 578 (delta 64), reused 94 (delta 43), pack-reused 441 (from 2)
Receiving objects: 100% (578/578), 29.08 MiB | 21.88 MiB/s, done.
Resolving deltas: 100% (290/290), done.
pi@TEST-MAG: ~ $ cd ~/UKRAA_Magnetometer/install
pi@TEST-MAG: ~/UKRAA_Magnetometer/install $ chmod +x *.sh
```

This will make the **install.sh** script executable.

3. Type the following command and press **enter**

sudo bash install.sh



```
pi@TEST-MAG: ~/UKRAA_Magnetometer/install
File Edit Tabs Help
pi@TEST-MAG:~ $ git clone https://github.com/UKradioastro/UKRAA_Magnetometer.git
Cloning into 'UKRAA_Magnetometer'...
remote: Enumerating objects: 578, done.
remote: Counting objects: 100% (137/137), done.
remote: Compressing objects: 100% (94/94), done.
remote: Total 578 (delta 64), reused 94 (delta 43), pack-reused 441 (from 2)
Receiving objects: 100% (578/578), 29.08 MiB | 21.88 MiB/s, done.
Resolving deltas: 100% (290/290), done.
pi@TEST-MAG:~ $ cd ~/UKRAA_Magnetometer/install
pi@TEST-MAG:~/UKRAA_Magnetometer/install $ chmod +x *.sh
pi@TEST-MAG:~/UKRAA_Magnetometer/install $ sudo bash install.sh
```

This will run the install script.

There may be occasions during the running of the install script that require you to make a keyboard entry.

When asked **Do you want to continue? [Y/n]** - type **Y** or **y** and press **enter**

That's it!

The code is now set up to run automatically; it will record the data from the PicoMagnetometer, process the data, plot the data and post the plots to your intranet web page.

There are other functions that are customisable by the user, such as **email alerts** and **FTP** to users' external website, that are covered in the **User Manual**.

Optional: run an install-time heartbeat smoke check (disabled by default):

sudo MAGNETOMETER_INSTALL_SMOKE_HEARTBEAT=1 bash install.sh

This runs **--test-heartbeat** once during install and prints a clear **HEARTBEAT_SMOKE_CHECK: PASS** or **HEARTBEAT_SMOKE_CHECK: FAIL** summary.

What does the code do?

Things it does right from the start

The code records data from the UKRAA PicoMagnetometer via serial over the supplied USB cable and stores the data to the raw data folder.

The raw data will be processed daily, via CRON, to get magnetic field values per minute in the X (N->S), Y (E->W) and Z (U->D) directions from your previous day's data. The maximum hourly variation of the magnetic field values in the X (N->S) and Y (E->W) directions will also be produced.

A number of plots will be created:

- X, Y and Z (North/South, East/West and up/down)
- Maximum hourly variation in X and Y directions

The raw data will also be processed on a continuous 5 minute basis, again via CRON, to generate a rolling 24 hour plot of X, Y and Z magnetic fields and % change of magnetic field for combined X and Y directions. The latter used to predict the potential of visible Aurora activity.

A simple web server and web page is set up on your RPi so that you can view your PicoMagnetometer results on your desktop PC and/or smart phone when connected to your home network. To access the webpage, see **PicoMagnetometer webpage** section for details.

Things it can do with selectable options

Additional plots

There is the option of the following additional rolling and daily plots

- H, D and Z (Local horizontal plane, declination angle and up/down)
- B and I (Total strength of Earth's magnetic field and angle of Earth's magnetic field)

These are configurable through **plot.ini** file, see **Optional HDZ and/or BI plots** section for details.

Email alerts

There is the option of receiving email alerts when an activity threshold is passed.

This is configurable through the **alerts.ini** file, see **Optional Rolling alert Emails** section for details.

Upload, via FTP, to external hosted website

There is the option of uploading all generated plot to an externally hosted webpage.

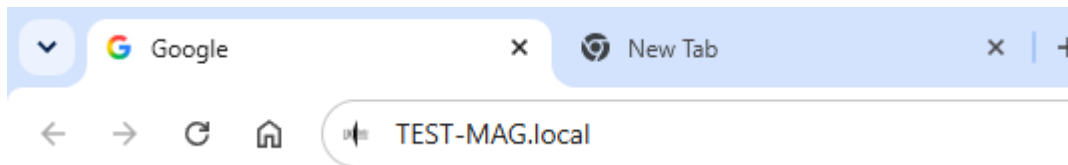
This is configurable through the **remote-upload.ini** file, see **Optional Remote FTP upload** section for details.

PicoMagnetometer webpage

To access the PicoMagnetometer webpage from your desktop PC or your smart phone...

1. Open your preferred web application (Safari, Chrome, Firefox, etc.).
2. In the search bar type the following and press enter

`http://test-mag.local`



This will take you to the web page for your PicoMagnetometer on your local intranet, displaying both the rolling 24 hour plots and yesterday's plots.

NOTE: if you have a different **hostname** for your RPi, change the search bar entry to...

<http://hostname.local>

Where **hostname** is the hostname for your RPi setup.

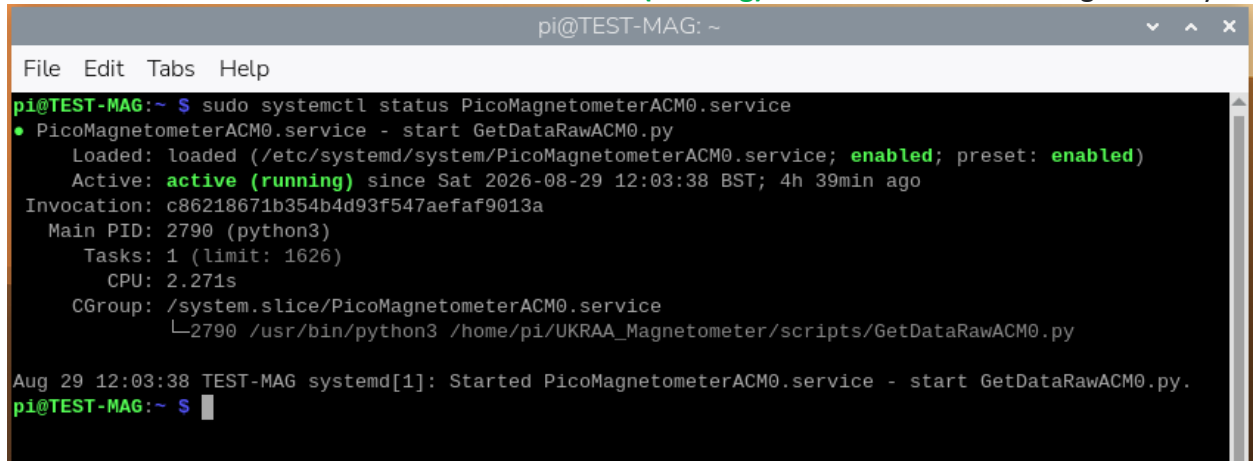
Code is also supplied that will enable the user to upload, via FTP, to their host website, should they have one - see **Optional remote FTP upload** section for more details.

Check PicoMagnetometerACM0 service is running

1. To check the **status** of your service, type the following command and press enter.

sudo systemctl status PicoMagnetometerACM0.service

Two **GREEN enabled** and a **GREEN active (running)** indicates service running correctly.



```
pi@TEST-MAG: ~  
File Edit Tabs Help  
pi@TEST-MAG:~$ sudo systemctl status PicoMagnetometerACM0.service  
• PicoMagnetometerACM0.service - start GetDataRawACM0.py  
   Loaded: loaded (/etc/systemd/system/PicoMagnetometerACM0.service; enabled; preset: enabled)  
   Active: active (running) since Sat 2026-08-29 12:03:38 BST; 4h 39min ago  
 Invocation: c86218671b354b4d93f547aefaf9013a  
    Main PID: 2790 (python3)  
       Tasks: 1 (limit: 1626)  
          CPU: 2.271s  
    CGroup: /system.slice/PicoMagnetometerACM0.service  
            └─2790 /usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/GetDataRawACM0.py  
  
Aug 29 12:03:38 TEST-MAG systemd[1]: Started PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
pi@TEST-MAG:~$
```

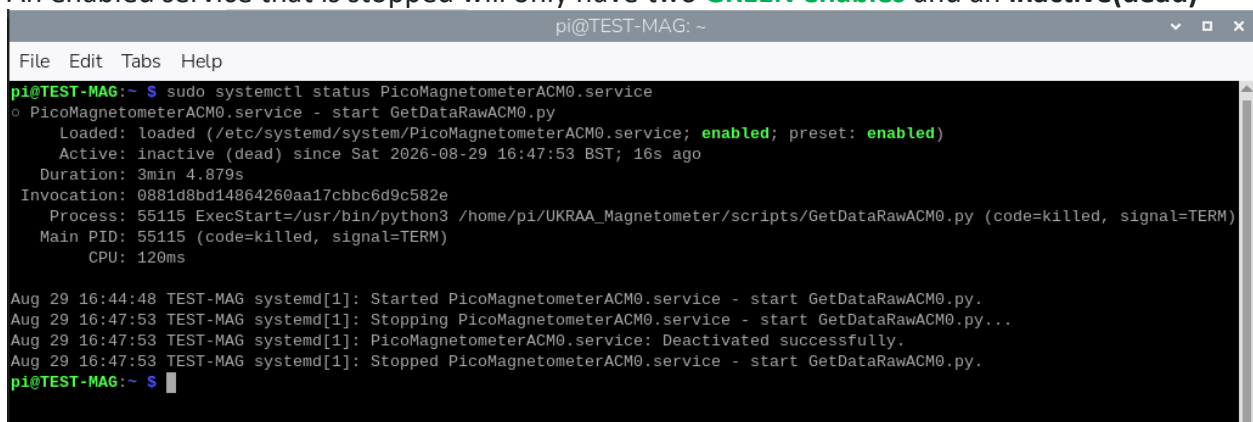
2. To **start** your service, type the following command and press enter.

sudo systemctl start PicoMagnetometerACM0.service

3. To **stop** your service, type the following command and press enter.

sudo systemctl stop PicoMagnetometerACM0.service

An enabled service that is stopped will only have two **GREEN enables** and an **inactive(dead)**



```
pi@TEST-MAG: ~  
File Edit Tabs Help  
pi@TEST-MAG:~$ sudo systemctl status PicoMagnetometerACM0.service  
• PicoMagnetometerACM0.service - start GetDataRawACM0.py  
   Loaded: loaded (/etc/systemd/system/PicoMagnetometerACM0.service; enabled; preset: enabled)  
   Active: inactive (dead) since Sat 2026-08-29 16:47:53 BST; 16s ago  
 Duration: 3min 4.879s  
 Invocation: 0881d8bd14864260aa17cbbc6d9c582e  
    Process: 55115 ExecStart=/usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/GetDataRawACM0.py (code=killed, signal=TERM)  
    Main PID: 55115 (code=killed, signal=TERM)  
       CPU: 120ms  
  
Aug 29 16:44:48 TEST-MAG systemd[1]: Started PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
Aug 29 16:47:53 TEST-MAG systemd[1]: Stopping PicoMagnetometerACM0.service - start GetDataRawACM0.py...  
Aug 29 16:47:53 TEST-MAG systemd[1]: PicoMagnetometerACM0.service: Deactivated successfully.  
Aug 29 16:47:53 TEST-MAG systemd[1]: Stopped PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
pi@TEST-MAG:~$
```

If this is the case you will need to:

- Start the service

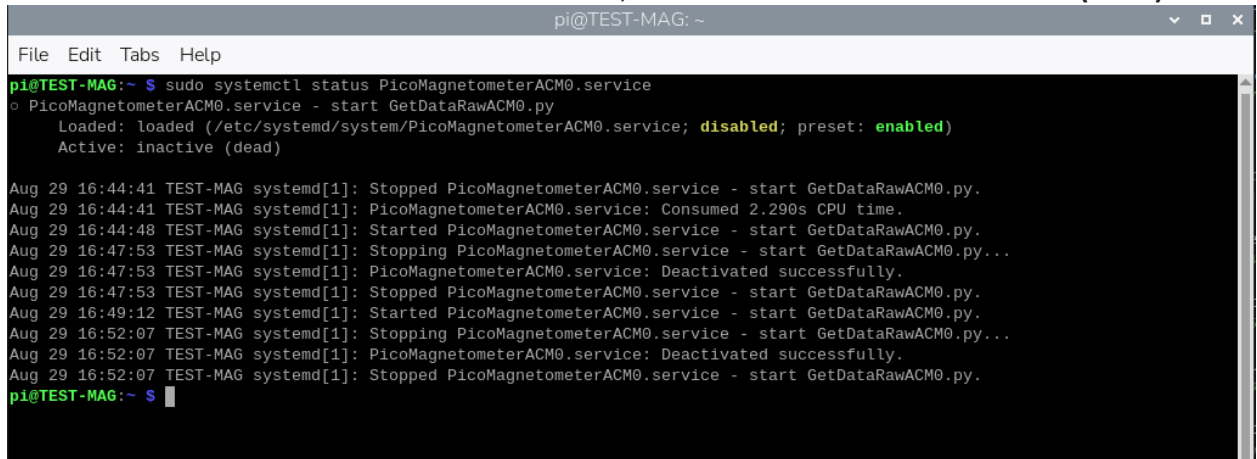
4. To **enable** your service, type the following command and press enter.

sudo systemctl enable PicoMagnetometerACM0.service

5. To **disable** your service, type the following command and press enter.

sudo systemctl disable PicoMagnetometerACM0.service

A disabled service will have a **YELLOW disabled**, a **GREEN enabled** and an **inactive (dead)**



```
pi@TEST-MAG: ~  
File Edit Tabs Help  
pi@TEST-MAG:~$ sudo systemctl status PicoMagnetometerACM0.service  
PicoMagnetometerACM0.service - start GetDataRawACM0.py  
Loaded: loaded (/etc/systemd/system/PicoMagnetometerACM0.service; disabled; preset: enabled)  
Active: inactive (dead)  
  
Aug 29 16:44:41 TEST-MAG systemd[1]: Stopped PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
Aug 29 16:44:41 TEST-MAG systemd[1]: PicoMagnetometerACM0.service: Consumed 2.290s CPU time.  
Aug 29 16:44:48 TEST-MAG systemd[1]: Started PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
Aug 29 16:47:53 TEST-MAG systemd[1]: Stopping PicoMagnetometerACM0.service - start GetDataRawACM0.py...  
Aug 29 16:47:53 TEST-MAG systemd[1]: PicoMagnetometerACM0.service: Deactivated successfully.  
Aug 29 16:47:53 TEST-MAG systemd[1]: Stopped PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
Aug 29 16:49:12 TEST-MAG systemd[1]: Started PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
Aug 29 16:52:07 TEST-MAG systemd[1]: Stopping PicoMagnetometerACM0.service - start GetDataRawACM0.py...  
Aug 29 16:52:07 TEST-MAG systemd[1]: PicoMagnetometerACM0.service: Deactivated successfully.  
Aug 29 16:52:07 TEST-MAG systemd[1]: Stopped PicoMagnetometerACM0.service - start GetDataRawACM0.py.  
pi@TEST-MAG:~$
```

If this is the case you will need to:

- Enable the service
- Start the service

Optional Post install operational checks

runPostUpdateChecksACM0.sh

After installing/updating scripts on your RPi, you can run:

```
sudo bash /home/pi/UKRAA_Magnetometer/scripts/runPostUpdateChecksACM0.sh
```

This executes:

- optional daily publish logic checks
- optional HDZ/BI web visibility checks

checkDailyPublishHealthACM0.sh

For daily runtime monitoring, a health marker is written by:

```
sudo bash /home/pi/UKRAA_Magnetometer/scripts/checkDailyPublishHealthACM0.sh
```

Marker output file:

```
/home/pi/UKRAA_Magnetometer/data/status/daily-health.txt
```

The marker reports one of three states:

- **PASS** – yesterday was processed and all expected plots are published.
- **PENDING** – no raw data was recorded for yesterday, so there was nothing to process. Expected on the first day after a fresh install, or if the acquisition service was stopped all day. Exit code 0.
- **FAIL** – raw data existed but the processed file or one or more published plots are missing. This is a real fault. Exit code 1.

showDashboardSummaryACM0.sh

One-line dashboard summary on demand:

```
sudo bash /home/pi/UKRAA_Magnetometer/scripts/showDashboardSummaryACM0.sh
```

If scheduled in cron, summary snapshots can be collected in:

```
/home/pi/UKRAA_Magnetometer/logfiles/dashboard-summary.log
```

Optional HDZ and/or BI plots

Within the `/home/pi/UKRAA_Magnetometer/config` folder there is a file named **plot.ini**.

By default the option of producing HDZ and/or BI plots is turned off (**false**), and the plots do not appear as an option from the intranet webpage.

Should the user wish to have access to HDZ and/or BI rolling plots and yesterday's plots for these options then change the following in the `plots.ini` file:

- For HDZ plots change line 13
 - From **plot_hdz = false**
 - To **plot_hdz = true**
- For BI plots change line 19
 - From **plot_bi = false**
 - To **plot_bi = true**

These plots will automatically be added to the intranet webpage if selected; with the rolling plot being present after 5 minutes and yesterdays plot being present after 09:30 the next day.

Optional Rolling alert Emails

Within the `/home/pi/UKRAA_Magnetometer/config` folder there is a file named **alert.ini**.

By default the option of producing rolling email alerts is turned off (**false**).

Should the user wish to receive rolling email alerts from their PicoMagnetometer follow the instructions below;

Rolling alert evaluation supports user defined activity thresholds.

Default values will be:

- **Yellow** at **50nT**
- **Amber** at **100nT**
- **Red** at **200nT**

The user can define their own alert values by modifying lines 18, 19 and 20 of **alert.ini**.

The alert evaluator runs as part of **processRollingData.sh** and only sends email on threshold transitions to levels you choose.

The same threshold values are also used by:

- Daily hourly Activity plot (**PlotDataActivityACM0.gp**)
- Rolling Activity plot (**PlotRollingActivityACM0.gp**)

Configure via alerts.ini file (recommended)

1. Copy the example file: **install/alerts.ini.example**
2. Place it on the Pi as: **/home/pi/UKRAA_Magnetometer/config/alerts.ini**
3. Edit the values for your SMTP service and recipients.
DO THIS BEFORE CHANGING THE STATE OF email_enabled
4. Set activity thresholds in the same file under **[alerts]**:

```
[alerts]
yellow_threshold_nt = 50
amber_threshold_nt = 100
red_threshold_nt = 200
```

Enable or disable alert emails

Email sending is off by default so a fresh install does not log SMTP failures every 5 minutes.

In **alerts.ini**:

```
[alerts]
email_enabled = true
```

Notes:

- When **email_enabled = false**, thresholds, alert levels, plots and the web status page all still work; only email sending is skipped.
- Suppression is logged once per level transition, not on every run.
- The daily heartbeat email is also skipped while **email_enabled = false**.
- The manual test scripts (**testAlertEmailACM0.sh**, **testHeartbeatEmailACM0.sh**) still send, so you can verify SMTP before enabling.
- Environment variable override: **MAGNETOMETER_EMAIL_ENABLED**

By default, **EvaluateAlertsACM0.py** reads: **/home/pi/UKRAA_Magnetometer/config/alerts.ini**

You can override the config path with: **MAGNETOMETER_ALERTS_INI_PATH=/path/to/alerts.ini**

Configure which levels trigger email

Set **MAGNETOMETER_EMAIL_ALERT_LEVELS** as a comma-separated list:

- **RED**
- **RED,AMBER**
- **RED,AMBER,YELLOW**

For instance:

- if you only require **RED** trigger alerts set **email_alert_levels = RED**
- if you require both **RED** and **AMBER** trigger alerts set **email_alert_levels = RED,AMBER**
- if you require all alerts set **email_alert_levels = RED,AMBER,YELLOW**

Configure SMTP and recipients

You can set SMTP/email values in the **alerts.ini** file, or by environment variables. Environment variables take precedence if both are set.

In **alerts.ini**:

```
[smtp]
host = smtp.example.com
port = 587
username = your-smtp-username
password = your-smtp-password
starttls = true
ssl = false

[email]
from = magnetometer@example.com
to = you@example.com
attach_plot = false
```

Notes:

- If **RollingActivity.png** exists, it is attached to the alert email if **attach_plot = true**

Available environment variables are:

- **MAGNETOMETER_SMTP_HOST** (required)
- **MAGNETOMETER_SMTP_PORT** (default 587)
- **MAGNETOMETER_SMTP_USERNAME** (optional)
- **MAGNETOMETER_SMTP_PASSWORD** (optional)
- **MAGNETOMETER_SMTP_STARTTLS** (*true* by default)
- **MAGNETOMETER_SMTP_SSL** (*false* by default)
- **MAGNETOMETER_EMAIL_FROM** (required)
- **MAGNETOMETER_EMAIL_TO** (required, comma-separated for multiple recipients)
- **MAGNETOMETER_EMAIL_ATTACH_PLOT** (*true* by default)
- **MAGNETOMETER_WEB_URL** (optional, included in email body)

Notes:

- Alert state is stored in **data/alerts/alert-state.json**
- Rolling status is read from **data/status/current.json**
- Alert evaluator config precedence is: environment variable -> **.ini** file -> built-in default

Regenerate a chosen Activity date (quick local test mode)

You can regenerate the hourly Activity plot for any date without storing an archive copy by default:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testActivityPlotACM0.sh YYYY-MM-DD
```

This updates only:

- **/home/pi/UKRAA_Magnetometer/temp/Activity.png**

To also write the dated archive file in **plots/Activity/**, add **--archive**:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testActivityPlotACM0.sh YYYY-MM-DD --archive
```

Send a one-off SMTP test email

To verify SMTP settings without waiting for a threshold transition, run:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testAlertEmailACM0.sh
```

Or run the Python script directly:

```
/usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/EvaluateAlertsACM0.py --test-email
```

The test email does not update transition state, so normal alert logic is unaffected.

Optional daily heartbeat email

You can enable a once-per-day summary email so you know the system is alive even when no threshold transition occurs.

In **alerts.ini**:

```
[heartbeat]
enabled = true
hour_utc = 9
to =
attach_plot = false
```

Notes:

- **hour_utc** is the first UTC hour of the day when the heartbeat can send.
- Only one heartbeat attempt is made per UTC day (success or failure), to avoid retry spam every 5 minutes.
- If **RollingActivity.png** exists, it is attached to the alert email if **attach_plot = true**
- If **heartbeat.to** is blank, it uses **[email] to**.

Environment variable overrides are also available:

- **MAGNETOMETER_HEARTBEAT_ENABLED**
- **MAGNETOMETER_HEARTBEAT_HOUR_UTC**
- **MAGNETOMETER_HEARTBEAT_TO**
- **MAGNETOMETER_HEARTBEAT_ATTACH_PLOT**

Send a one-off heartbeat test email immediately

To verify heartbeat email delivery without waiting for **heartbeat.hour_utc**, run:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testHeartbeatEmailACM0.sh
```

Or run the Python script directly:

```
/usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/EvaluateAlertsACM0.py --test-heartbeat
```

This immediate heartbeat test does not update daily heartbeat schedule/state tracking.

Optional Remote FTP upload

Within the `/home/pi/UKRAA_Magnetometer/config` folder there is a file named **remote-upload.ini**.

By default the option of uploading plot PNG files is turned off (**false**).

Should the user wish to upload plot PNG files from their PicoMagnetometer to an externally hosted webpage follow the instructions below;

You can upload plot PNG files to an external hosting site while keeping local web publishing as the primary path.

Remote upload config file:

- `/home/pi/UKRAA_Magnetometer/config/remote-upload.ini`

Template created by installer:

- `install/remote-upload.ini.example`

Example settings:

```
[ftp]
enabled = true
site = your-uploader-fp
user = your-upload-user
password = your-upload-password
port = 21
directory = /data
timeout_seconds = 30
passive = true
create_dirs = true
upload_status_json = false
```

Behaviour:

- Daily run (09:30 via `moveGraphs.sh`) uploads:
 - **Activity.png, XYZ.png, HDZ.png, BI.png** to `/data`
- Rolling run (every 5 minutes via `processRollingData.sh`) uploads:
 - **RollingActivity.png, RollingXYZ.png, RollingHDZ.png, RollingBI.png** to `/data/rolling`
- Optional rolling status JSON upload:
 - set **upload_status_json = true**
 - uploads **data/status/current.json** to `/data/status/current.json`
 - external webpage status fallback (**status/current.json**) requires this to be true.
- Remote upload failures are non-blocking and do not stop local publishing.

Manual test commands:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/uploadRemoteACM0.sh daily  
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/uploadRemoteACM0.sh rolling
```

Combined test helper:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testRemoteUploadACM0.sh
```

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Contact us

Please send an e-mail to PicoMagnetometer@ukraa.com

